

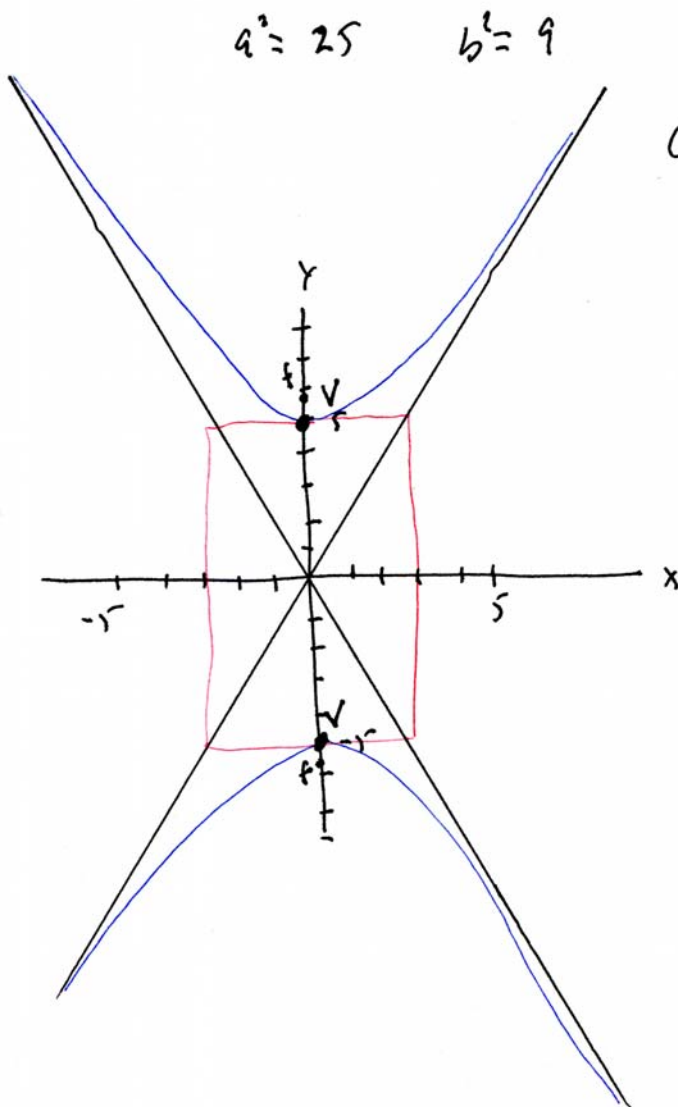
Name _____

Math 1350, Quiz #6

1. (Section 5.3, Problem 17) Find the equation of the ellipse with foci $(\pm 6, 0)$ and eccentricity $e = 3/5$.

$$\begin{aligned} e &= \frac{c}{a} & c &= 6 & \frac{3}{5} &= \frac{6}{a} & a &= 10 \\ c^2 &= a^2 - b^2 & b^2 &= a^2 - c^2 = 100 - 36 = 64 & b &= 8 \\ \frac{X^2}{10^2} + \frac{Y^2}{8^2} &= 1 \end{aligned}$$

2. (Section 5.4, Problem 7) Sketch and discuss the graph of $\frac{y^2}{25} - \frac{x^2}{9} = 1$. Specifically, identify: vertices, foci, transverse axis, conjugate axis, equations of the asymptotes, and the eccentricity.



$$V (0, \pm 5)$$

$$F (0, \pm \sqrt{34})$$

transverse y-axis

conjugate x-axis

$$\frac{y}{5} = \pm \frac{x}{3}$$

$$e = \frac{c}{a} = \frac{\sqrt{34}}{5}$$